MILITARY SPECIFICATION

CARTRIDGE, 81MM SMOKE, RED PHOSPHORUS M819, WITH FUZE MTSQ M772 LOADING ASSEMBLING AND PACKING

This Specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for the loading, assembling, and packaging of one type of mortar cartridge designated as Cartridge, 81MM Smoke, Red Phosphorus (R.P.), M819.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286	-	Propellants, Solids, Sampling, Examination and Testing
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1234	-	Pyrotechnics: Sampling, Inspection and Testing

2.1.2 Other government documents, drawings and publications. The following other documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.3)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8796521	-	Marking and Sealing for Ammunition Fiber Containers
8796522	-	Marking Diagram and Sealing for Wood Packing Boxes
9327839	-	Cartridge, 81MM Smoke, R.P. M819 w/Fuze Time, M772
9362798	-	Packing and Marking for box, Ammunition, Packing for Cartridge, 81MM, M819 and M853
9362799	-	Packing and Marking for Container, Ammunition Fiber for Cartridge, 81MM, M819 and M853

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings, specifications, standards and this specification.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9327839 and associated dwgs., and with all requirements specified in applicable specifications and standards. Components shall also be free of burrs, chips, dirt, corrosion and other foreign material that may adversely affect fit or function. The cleaning method used shall not be injurious to the parts, nor shall any parts be contaminated.

3.3 Strength of tack weld. The tack weld shall meet the pull test requirements specified on drawing 9327838 when tested in accordance with 4.5.4.

3.4 Proving ground test. The cartridge shall comply with the following requirements when tested as specified in 4.5.5.

3.4.1 Misfires. The cartridge shall not fail to fire from the weapon.

3.4.2 Fuze/separation. The fuze shall function and the tail cone assembly shall separate from the body tube assembly in flight.

3.4.3 Smoke production. All pellets shall be fuze ignited and produce smoke on the ground after being ejected in flight by proper functioning of the fuze. Pellets shall survive functioning of the fuze without shattering.

3.4.4 Security of parts. There shall be no metal parts failure in the bore of the weapon or in flight.

3.4.5 Horizontal range. No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

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3.4.6 Fuze function time. The fuzes shall function and shall comply with the fuze function time in Table I.

TABLE I

Fuze Setting (Sec)	Charge	Fuze Function Time (Sec)
13.5	1	13.5 ± 1.5
33.5	2	33.5 ± 1.5

3.4.7 Premature burst. The fuze shall not function in the bore of the weapon or in the first 100 meters of flight.

3.5 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2).

3.6 Workmanship.

3.6.1 Parts and assemblies. Parts and assemblies shall be free from gouges, dents, cracks, deformation or any similar defects.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. No part shall have a burr which might be injurious to personnel handling the item.

3.6.4 Painting and markings. Markings shall be accurate, neat, complete and legible. Paint shall be smooth and dry when inspected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.1.1 Responsibility for compliance. All items must meet all requirements in sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated (See Table I).

Part description	Drawings	Quantity
Obturating ring	9310339	10
Fiber container, untaped	9354333	5
Body tube assembly	9327832	5
Projectile loading assembly	9327838	5
M819 cartridge with M772 fuze in taped fiber container	9362899	5

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	SHEET 1 OF 1 / DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge, M819, Components and Assemblies				See Below
	Obturating Ring (Dwg. 9310339)				
	Strength of tack weld test	10	100%	3.3	4.5.4
	Projectile Loading Assembly (Dwg. 9327838)				
	Examination for defects	5	100%	3.2	4.4.2.5
	Unsealed Fiber Container (Dwg. 9354333)				
	Examination for defects	5	100%	3.2	4.4.2.9
	Ammunition Container after Sealing of Cartridge (Dwg. 9362899)				
	Examination for defects	5	100%	3.2	4.4.2.10
	Cartridge, 81MM Smoke RP M819 Dwg. 9327839				
	Examination for defects	5	100%	3.2	4.4.2.7
	Ballistic test (proving ground)	5	100%	3.4	4.5.5

NOTES:

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4.4. Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of Cartridges, 81MM, Smoke, RP, M819 shall contain:

a. Projectile body metal parts from one lot interfix number, from one manufacturer.

b. Ignition cartridges, of one lot interfix number, from one manufacturer.

c. Fin assemblies, of one lot interfix number, from one manufacturer.

d. Propelling charges, made from not more than one lot of propellant, from one manufacturer.

e. Loaded fuzes of one lot interfix number from one manufacturer.

f. Red phosphorus pellets from one lot interfix number from one manufacturer.

g. Obturating rings, of one lot interfix number, from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Body Tube Assembly				9327832
					NEXT HIGHER ASSEMBLY 9327838
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
102	Protective finish present on rear inside diameter (I.D.) or adapter threads		0.40%	3.2	Visual
103	Threads damaged		0.40%	3.2	Visual
104	Burrs or foreign matter in threads		0.40%	3.2	Visual
Minor					
201	Protective finish missing or not full		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70549 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Projectile, 81MM Smoke, RP M819 Payload and Loading Assembly, Prior to Tail Cone Assembly	1 OF 1	9327838
			NEXT HIGHER ASSEMBLY: 9327839

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Pellet(s) cracked or broken		0.40%	3.2	Visual
102	Felt spacer(s) missing		0.40%	3.2	Visual
103	Cardboard overlapped or missing		0.40%	3.2	Visual
104	Forward felt spacer missing		0.40%	3.2	Visual
105	Piston plate missing or damaged		0.40%	3.2	Visual
106	Payload stack height incorrect		100%	3.2	Gage 1/ See 4.5.3
107	Rear felt spacer missing		0.40%	3.2	Visual
108	Foil missing from piston plate or facing incorrectly		0.40%	3.2	Visual
109	Silastic seal ring missing		0.40%	3.2	Visual
Minor					
201	Tape missing (if applicable)		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: 1/ If uploading payload followed by insertion and pinning of tailcone is automated and continuous, inspection must be automated.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Tail Cone and Rear Support Disc Assembly	1 OF 1	9300233
			NEXT HIGHER ASSEMBLY 9327838

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Rear support disc cracked		100%	3.1	Visual
Major					
101	Rear support threads damaged		0.40%	3.2	Visual
102	Sealant missing or inadequate		0.40%	3.2	Visual
103	Tail cone threads damaged		0.40%	3.2	Visual
104	Rear support disc not fully seated		0.40%	3.2	Gage
105	Tack weld strength inadequate	1/5	100%	3.3	Gage/4.3.4.5
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: 1/ 5 consecutive assemblies shall pass this test whenever equipment is started adjusted, or left idle for more than 1 hour.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70549 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630574
4.4.2.4	Tape, Aluminum Foil				NEXT HIGHER ASSEMBLY 12630566
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of foil		0.40%	3.2	Gage
102	Diameter of foil		0.40%	3.2	Gage
103	Foil punctured		0.40%	3.2	Visual
Minor					
201	Evidence of rough handling		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Projectile, 81MM Smoke, RP M819 Loading Assembly, Including Tail Cone Assembly				9327838
					NEXT HIGHER ASSEMBLY 9327839
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Support plate missing	1/	100%	3.2	Gage
Special					
a	Warning missing, incomplete or incorrect		100%	3.2	Visual
Major					
101	Sealant missing or inadequate at tube/ tail cone interface		0.40%	3.2	Visual
102	Depth of shear pin holes	2/		3.2	Gage
103	Location of shear pin holes	3/	0.40%	3.2	Gage
104	Diameter of shear pin holes	2/		3.2	Gage
105	Shear pin(s) missing, rear		0.40%	3.2	Visual
106	Shear pin(s) not flush or below		0.40%	3.2	Gage
107	Protective dust plug missing (if applicable)		0.40%	3.2	Visual
108	Obturator band missing		100%	3.2	Visual
109	Obturator not flush or below		0.40%	3.2	Visual
110	Tack weld missing or improperly positioned		0.40%	3.2	Visual
111	Tail cone not fully seated		0.40%	3.2	Gage

NOTES: 1/ If insertion and pinning of tail cone assembly is automated and continuous with assembly of cartridge, inspection must be automated. 2/ Five consecutive projectiles must meet the requirement each time a drill bit is changed or repositioned. The shear pins shall not be inserted into the five sample units. 3/ Tool control acceptable. Check at the start of each shift and after applicable tool change or adjustment.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Projectile, 81MM Smoke, RP M819 Loading Assembling, Including Tail Cone Assembly (Cont)	2 OF 2	9327838
			NEXT HIGHER ASSEMBLY 9327839

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Foil tape bubbled wrinkled punctured, missing or not covering piston plate hole		0.40	3.2	Visual
113	Tape not fully applied		0.40%	3.2	Visual
Minor					
201	Obturating band incorrectly assembled (reversed split)		0.65%	3.2	Visual
202	Protective dust plug missing		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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MIL-C-70549 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Ring, Silastic Seal	1 OF 1	12630575
			NEXT HIGHER ASSEMBLY: 9327839

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.2	Gage
102	Thickness of ring		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
104	Ring damaged		0.40%	3.2	Visual
105	Evidence of deterioration		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Cartridge, 81MM Smoke, RP, M819, with Fuze, Time, M772				9327839
					NEXT HIGHER ASSEMBLY 9362899
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Fuze foil sealing disc missing, punctured or not covering all holes		100%	3.2	Visual
Special					
a	Ignition cartridge cracked or damaged		100%	3.2	Visual
b.	Propelling charge(s) cracked or damaged		100%	3.2	Visual
c.	Fuze safety wire missing		100%	3.2	Visual
d.	Fuze not set on "s" (shipping) position		100%	3.2	Visual
e.	Silastic ring missing		100%	3.2	Visual
Major					
101	Damaged threads on fin boom, tail cone, fuze or fuze adapter, prior to assembly		0.40%	3.2	Visual
102	Adhesive missing or inadequate on mating surfaces of fin boom/tail cone and fuze/fuze adapter threads		0.40%	3.2	Visual
103	Assembly fails to freely traverse alignment gage		100%	3.2	Gage
104	Damaged threads on ignition cartridge or rear of fin boom		0.40%	3.2	Visual
105	Adhesive missing or inadequate on threads of ignition cartridge/fin boom interface		0.40%	3.2	Visual
106	Inadequate torque applied to fin boom	1/	100%	3.2	Torque test/4.5.2

NOTES:

1/ 25 fin assemblies shall be randomly checked after the start of each shift.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Cartridge, 81MM Smoke, RP M819, with Fuze, Time, M772 (Cont)	2 OF 2	9327839
			NEXT HIGHER ASSEMBLY 9362899

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
107	Security of parts		0.40%	3.2	Manual
108	Damaged fin blades		0.40%	3.2	Visual
109	Diameter over max, after paint touch-up	1/	100%	3.2	Gage
110	Propelling charge support missing or improperly assembled		0.40%	3.2	Visual
111	Foil tape not centered on back of fuze		0.40%	3.2	Visual
112	Ballistic test (proving ground)	50		3.4	4.5.5
Minor					
201	Protective finish damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: 1/ Only items with touch up paint to be checked. Repaint and recheck if the ring removes dried paint.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Container, Ammunition, Fiber, for Cartridges M819/M853 Before Securing				9354333
					NEXT HIGHER ASSEMBLY
					9362899/9362798
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Parts not secure (inside cap or container)		0.40%	3.2	Manual
102	Container damaged		0.40%	3.2	Visual
103	Absence of gap between cap and tube prior to taping		0.40%	3.2	Visual/5.1
Minor					
201	Markings incorrect, illegible or missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Container, Ammunition Fiber for Cartridges M819/M853 After Securing	1 OF 1	93262899/8796521
			NEXT HIGHER ASSEMBLY: 9354616

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Cartridge loose inside container		0.40%	3.2	Manual
102	Inner sealing strip not applied smoothly or oriented improperly		0.40%	3.2	Visual
103	Outer sealing strip not applied smoothly or oriented improperly		0.40%	3.2	Visual
104	Outer tape not folded into tab at end		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70549 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Box, Wirebound, Packing, Ammo for 81MM Mortar Prior to Closing				9354616
					NEXT HIGHER ASSEMBLY
					9362798/8796522
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Containers not firmly packed in box		0.40%	3.2	Manual
102	Inside length, width or depth inadequate		0.40%	3.2	Gage
Minor					
201	Box or nails loose		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Box, Wirebound, Packing, Ammo for 81MM Mortar After Closing				9362798/8796522
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Contents of box loose		0.40%	3.2	Manual
102	Markings missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.5 Proving ground. Fifty sample cartridges shall be selected for this test. The test shall be performed as specified in 4.5.5. Defect classification and individual accept/reject criteria shall be in accordance with the following table:

TABLE II

Categories	Defect	Accept/Reject	
Critical:	Premature burst or metal parts separation	0	1
Major:	Misfire	1	2
Major:	Fuze non-function or non-separation of tail cone from body tube assembly	1	2
Major:	Horizontal range less than 80% of average range	1	2
Major:	Fuze functioning time (See Note 1)	2	3
Major:	Payload fails to ignite or provide smoke	1	2

Note 1: If the fuze performance of any round in the sample exceeds the timing requirements specified, and the round functions point detonating, the fuze is considered a non-function.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to

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determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Torque test. The samples supplied in accordance with 4.2.3.4 shall be tested by applying the required assembly torque to the components in the direction of assembly before the thread sealing compound has set. Any perceptible movement of parts will be considered as failure of the assembly to comply with the drawing requirement.

4.5.3 Payload stack height. In order to avoid measurement error due to compressibility of felt spacers, it is recommended that a 2.5 inch diameter plate, 0.75 inches thick weighing approximately one pound should be placed on the top of the payload. Distance shall then be measured from the top of the plate to the rear face of the body tube assembly.

4.5.4 Tack weld strength. Tack welds shall be tested by assembling an obturating ring to a tail cone (or equivalent fixture) which has been modified to accomodate a hook which is used to pull the ring radially outward from underneath. The tail cone shall be held securely during the test. The force shall be applied gradually by mechanical means and shall be measured by a scale which indicates the highest value achieved prior to breakage. Failure of any tack weld to meet the requirement in 3.3 shall be cause for rejection. In performing the test, the tack weld shall be positioned 180° away from the hook used to pull on the obturator.

4.5.5 Proving ground. See 4.4.5. The test shall be performed at a Government Proving Ground in accordance with applicable USA TECOM procedures. All the cartridges shall be fired from a ground mounted 81mm Mortar for which the round is standard. The cartridges shall be assembled with accepted fuzes, fin assemblies, projectile assemblies, ignition cartridges, primers, and propellant increments. One half of the sample cartridges shall be fired with one propelling charge increment (charge zone one) and one half of the sample cartridges shall be fired with four propelling charge increments (charge zone four) and in accordance with Table III. However, if conditions or time of fuze functioning so dictate, the proving ground shall adjust the fuze in order to ensure airburst. Observation shall be made for compliance with the requirements specified for security of parts, functioning time, horizontal range

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and smoke production. Fuze functioning time shall be reported to at least the nearest 0.01 second, and horizontal range on the ground to a point under the position of fuze functioning to the nearest 5 meters. The cartridges shall be conditioned at 70 plus or minus 2.5 degrees Fahrenheit for a minimum of 4 hours prior to testing and fired at this temperature. Any malfunction occuring during the test shall be reported in the offical firing record. Meteorological data shall also be recorded for each round fired and reported in the offical firing record.

TABLE III

Charge Zone	Fuze Setting (seconds)	Mortar Elevation (degrees)
1	13.5	45
4	33.5	45

4.5.6 Security of parts. Parts to be inspected for security shall be manually checked to be tightly and securely attached as described in the applicable drawings. Such parts will not be readily loosened or evidence rattling or other apparent signs of poor assembly when so checked.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging of cartridges shall be in accordance with Dwg. 9362799 and marking in accordance with Dwg. 8796521. Prior to taping, the felt spacers shall cause a gap of approximately 1/8 inch between the cap and tube of the fiber container when unrestrained. Fiber containers shall be securely packed by tightly stuffing the overpacking. Secure box in accordance with Dwg. 9362798-1 and mark in acccordance with Dwg. 8796522.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for assembly into M819 smoke cartridges for use with 81mm mortars.

6.2 Ordering data. See MIL-A-48078.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.4 Submission of inspection equipment for design approvals. See para 6.2.3 of MIL-A-48078 (AR). Submit designs as required to Commander, US Army Armament Research, Development and Engineering Center, (ARDEC), ATTN: AMSMC-QAT-I (D), Dover, NJ 07801-5001.

6.5 Submission of test data. One copy of the proving ground ballistic acceptance test data (i.e., firing record) shall be forwarded to each of the following:

Commander
US Army Armament Research, Development and Engineering Center
ATTN: AMSMC-QAT-B(D)/SMCAR-AEE-WW(D)
Dover, NJ 07801-5001

6.6 Alternate procedures. Alternate procedures to those contained in this specification shall be approved in advance by: Commander, US Army Armament Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAT-B(D), Dover, NJ 07801-5001

6.7 Material safety data sheets. The contracting officer will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in Appendix B of FED-STD-313.

6.8 Subject term (key word) listing.

Ammunition
Cartridge, mortar
Misfire
Pellets, red phosphorus
Weld, tack

Custodian:
ARMY-AR

Preparing activity:
Army-AR

(Project 1315-A679)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70549	CARTRIDGE, 81MM SMOKE, RED PHOSPHORUS M19

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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